UIT THE ARCTIC UNIVERSITY OF NORWAY
FALCON 9 RE-ENTRY
TROMSØ, NORWAY

UiT MEMO #001
21 October 2026

To: Falcon 9 re-entry assessment
From: Juha-Pekka Vierinen
Subject: **Trajectory reconstruction and uncertainty**

TL;DR. We reconstruct the 19 February 2025 re-entry of Falcon 9 upper stage NORAD 62878 by connecting its final TLE state to the optically measured positions of fragments F_1 and F_2. The 3D position RMS is 1.81 km and 1.92 km, respectively. The best-fit paths and 128 uncertainty refits are available as a downloadable HDF5 data file.

The fragments were observed around 03:44–03:45 UTC. F_1 is most likely the rocket motor and F_2 the fuel tank assembly. The TLE state at its epoch, 02:13:45.787 UTC, and the timed dual-camera triangulated positions constrain the reconstruction. We compare it with ESA OrbGen, which used the TLE and ground-recovered debris positions as constraints. The optical observations and fragment identifications are described by Vierinen et al.[1]

The fragments share position and velocity at an inferred separation point, fitted here at 123.71 km altitude. Separation was not observed; no impulse is applied.

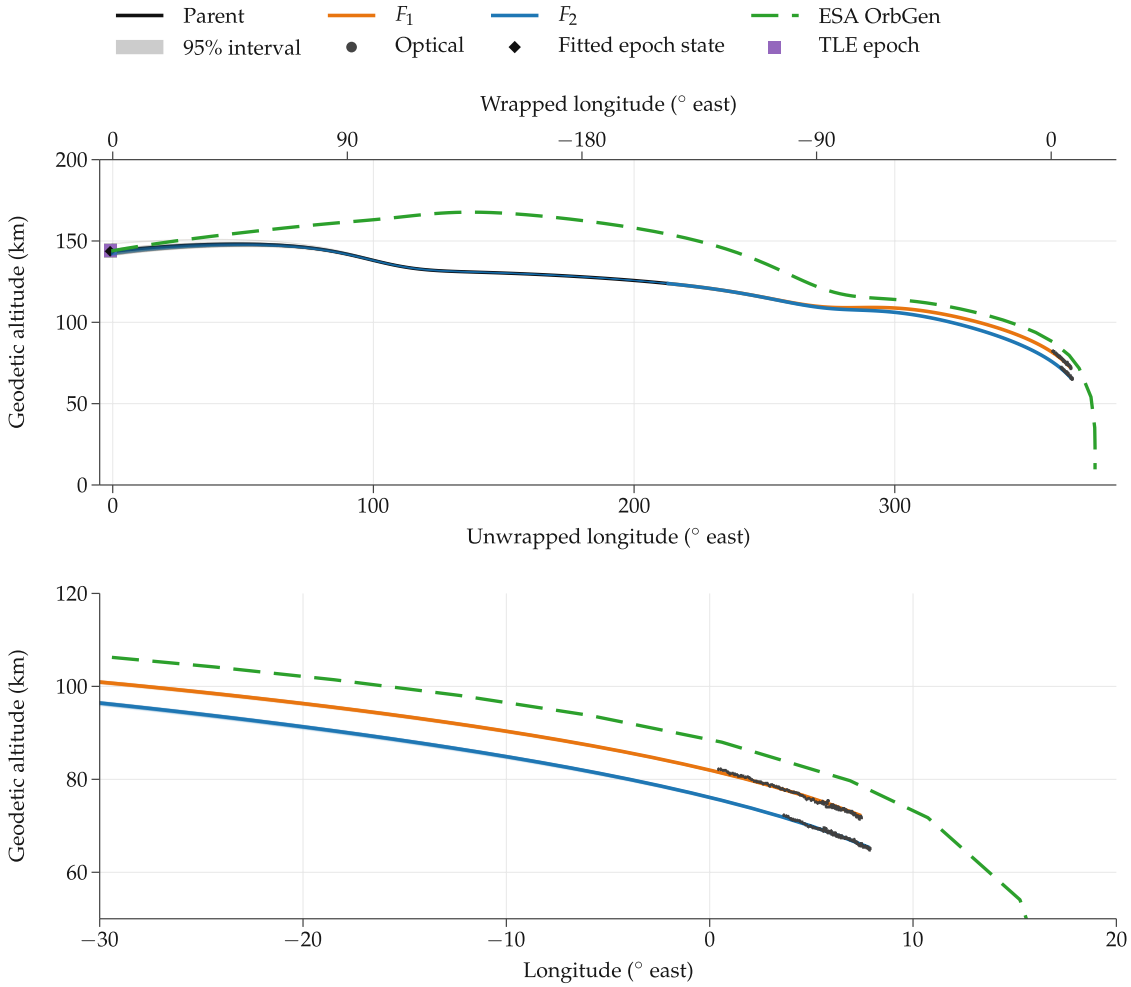


Figure 1: Reconstructed full arc (top) and atmospheric entry (bottom), with optical positions and the ESA OrbGen comparison. Shading shows the 95% conditional altitude interval at fixed longitude; faint lines show individual refits.

[1] Vierinen et al., *Optical and Radar Observations of the February 2025 Falcon 9 Upper-Stage Re-entry*, EGUsphere [preprint], 2026. https://doi.org/10.5194/egusphere-2026-2857

Model. The equation of motion is

$$\dot{\mathbf{r}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = \mathbf{g}(\mathbf{r}, t) - \tfrac{1}{2}\rho_{\mathrm{MSIS}}\,\beta_{\mathrm{eff}}(t)|\mathbf{v}_{\mathrm{rel}}|\mathbf{v}_{\mathrm{rel}}, \qquad \beta_{\mathrm{eff}} = \gamma C_D A/m. \tag{1}$$

Gravity includes EGM2008 to degree and order 180, the Sun and Moon. MSIS2.1 uses the observed solar and geomagnetic drivers; the atmosphere co-rotates with Earth. The effective drag coefficient is log-linear between its nodes: two parent endpoints, followed by independent outgoing and final values for each fragment. Position and velocity remain continuous at separation. Lift and winds are omitted. The numerical propagator is available at github.com/jvierine/numerical_propagator.

The initial-state priors have radial/along-track/cross-track position standard deviations of 100/400/100 km and velocity standard deviations of 100 m/s per component, truncated by $\|\mathbf{v}_{\mathrm{epoch}} - \mathbf{v}_{\mathrm{TLE}}\| \le 200$ m/s. These are chosen prior constraints, not a catalogue covariance. Each fragment has its own clock offset, bounded to ±1 s.

Fit quality and uncertainty.

Fragment	Positions	3D RMS (km)	Height RMS (km)
F_1	86	1.81	0.39
F_2	63	1.92	0.27

We use 128 bootstrap refits with optical errors shared within five-second blocks. The radial/along-track/cross-track error scales are 0.40/1.59/0.77 km for F_1 and 0.27/1.68/0.89 km for F_2. All trajectory parameters are refitted; the TLE prior centres and scales stay fixed. Every refit uses degree-180 gravity. The epoch-state corrections are 5.91 km (95% interval 3.97–8.91 km) and 19.46 m/s (19.10–20.52 m/s). The inferred separation altitude is 123.71 km (123.19–124.39 km). Shading gives pointwise 2.5–97.5 percentile intervals conditional on these errors, TLE constraints and drag model, rather than calibrated absolute orbit accuracy. The fitted coefficient is $\beta_{\mathrm{eff}} = \gamma C_D A/m$; density and area-to-mass effects cannot be determined separately.

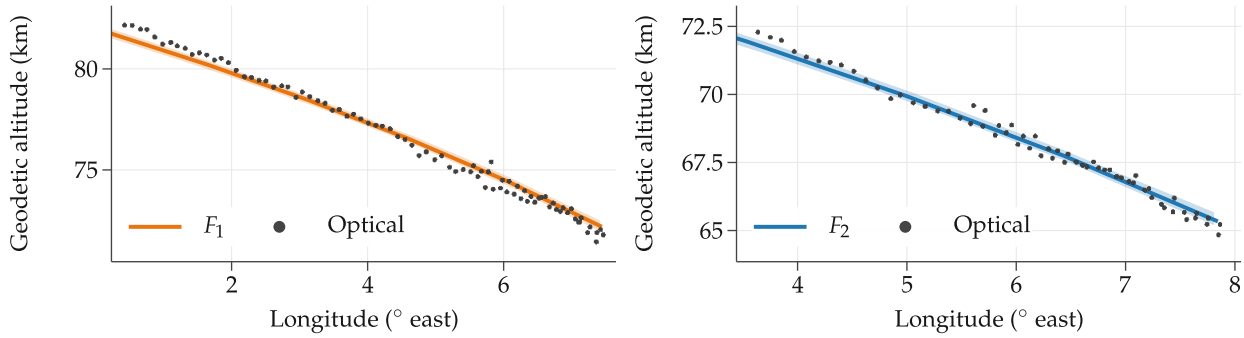


Figure 2: Optical positions and fitted fragment paths, with the same sampled 95% intervals as Figure 1.

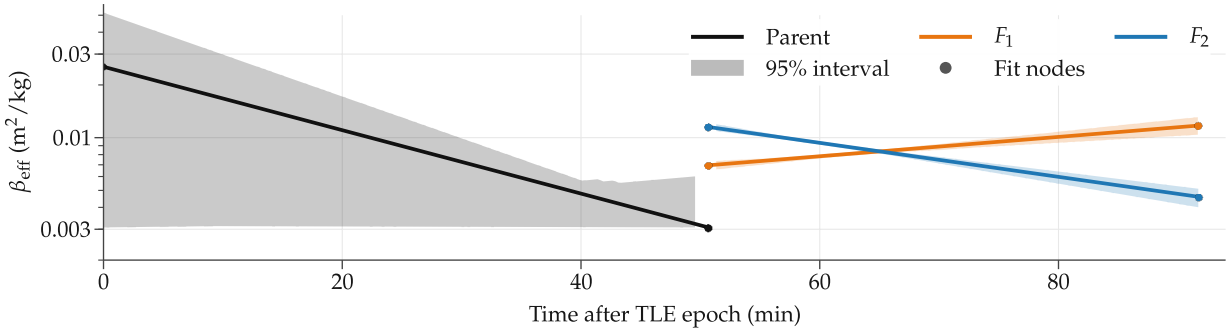


Figure 3: Effective $\gamma C_D A/m$ with conditional 95% intervals; circles mark the fitted endpoints. Parent drag is weakly constrained.

Persistent camera bias, other drag profiles, lift and winds are not covered. A positive Gaussian $\gamma \sim \mathcal{N}(1, 0.30^2)$ converts β_{eff} to physical $C_D A/m$ without being applied to the force a second time.